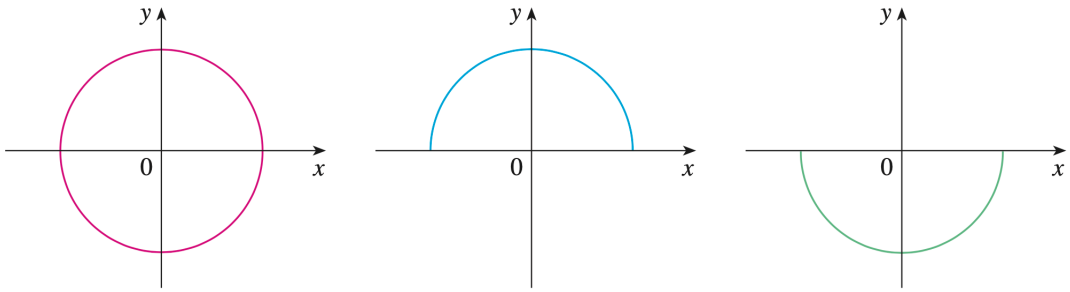
So far we have focused on calculating derivatives of functions of the form $y = f(x)$, where y is defined *explicitly* as a function of x. However, relations between variables are often expressed *implicitly*. For example, the equation of the circle $x^2 + y^2 = 25$ does not specify y directly, but rather defines y implicitly. This equation does not represent a single function because its graph fails the vertical line test. If, however, the equation $x^2 + y^2 = 25$ is solved for y, then *two* functions emerge: $y = -\sqrt{25 - x^2}$ and $y = \sqrt{25 - x^2}$.



(a) $x^2 + y^2 = 25$ (b) $f(x) = \sqrt{25 - x^2}$ (c) $g(x) = -\sqrt{25 - x^2}$

1.) We have identified two explicit functions that describe the circle. Find their derivatives using the Chain Rule.

If $y = \sqrt{25 - x^2}$, then $\dfrac{dy}{dx} =$ (1)

If $y = -\sqrt{25 - x^2}$, then $\dfrac{dy}{dx} =$ (2)

We use Equation (1) to find the slope of the tangent line at any point on the upper half of the unit circle and Equation (2) to find the slope of the tangent line at any point on the lower half of the circle.

2.) The equation $x - y^2 = 0$ implicitly defines what two functions? Use these two functions to find $\frac{dy}{dx}$ at any given point on the curve below.

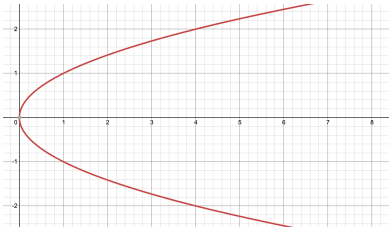
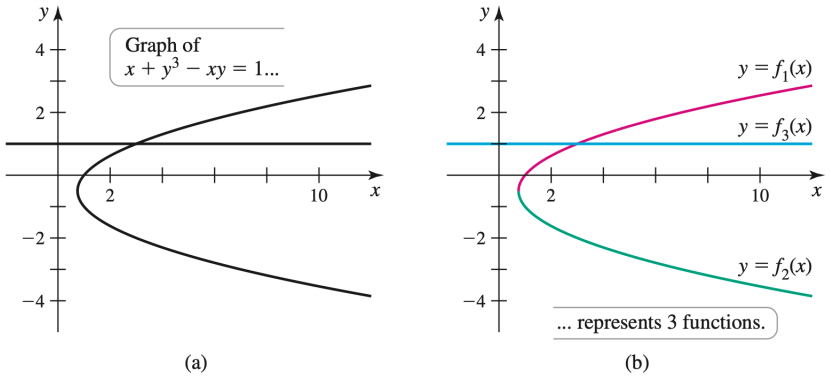


Figure 1: Graph of the solution set of the equation $x - y^2 = 0$.

While it is straightforward to solve some implicit equations for y (such as $x^2+y^2=1$ or $x-y^2=0$), it is difficult or impossible to solve other equations for y.

Example 1. Consider the equation $x+y^3-xy=1$. If we re-write this as $y^3-1=xy-x$, then using the difference of cubes formula $a^3-b^3=(a-b)(a^2+ab+b^2)$ we can factor each side as $(y-1)(y^2+y+1)=(y-1)x$. If $y=1$, then both sides are 0, so the equation holds for any value of x. Thus, all points of the form $(x,1)$ are solutions to this equation, which is the graph of the constant function $y=1$. If $y\neq 1$, then $(y-1)\neq 0$, so we may cancel this factor from both sides to obtain $y^2+y+1=x$. Subtracting x from both sides, this becomes $y^2+y+(1-x)=0$, and the quadratic formula gives two functions: $y=\frac{\sqrt{4x-3}-1}{2}$ and $y=-\frac{\sqrt{4x-3}-1}{2}$. The graph of the solution set to this equation is given below:



Using these functions, we find $\frac{dy}{dx}=\frac{1}{\sqrt{4x-3}}$ for each point on the top half-parabola except $(x,y)=(3,1)$, $\frac{dy}{dx}=0$ at $(3,1)$, and $\frac{dy}{dx}=-\frac{1}{\sqrt{4x-3}}$ for each point on the lower half-parabola.

Example 2. Consider the equation $y^5+xy=3$. It was proved in 1824 by Niels Henrik Abel that no general formula can be given for the roots of a fifth-degree polynomial equation in terms of radicals. (Later it was proved by Evariste Galois that it is impossible to find a general formula for the roots of an nth degree polynomial equation (in terms of algebraic operations on the coefficients) if n is any integer larger than 4.) For the equation above, the function $y=f(x)$ cannot be given explicitly. It is straightforward to check, e.g., that the point $(x,y)=(2,1)$ satisfies the equation, but how can we find $\frac{dy}{dx}$ at $(2,1)$? When we say that f is a function defined implicitly by $y^5+xy=3$, we mean that there exists a function $y=f(x)$ such that

$$(f(x))^5+xf(x)=3$$

for all values of x in the domain of f. Differentiating each side of this equation using the chain rule gives

$$5(f(x))^4f'(x)+f(x)+xf'(x)=0$$

We can solve this equation for $f'(x)$ to obtain

$$f'(x)=-\frac{f(x)}{5(f(x))^4+x}.$$

Since $y = f(x)$ and $\frac{dy}{dx} = f'(x)$, this equation says

$$\frac{dy}{dx} = -\frac{y}{5y^4 + x}.$$

Thus, given a point (x, y) on the curve, we can calculate $\frac{dy}{dx}$ in terms of (x, y). For example, at $(2, 1)$,

$$\frac{dy}{dx} = -\frac{1}{5(1)^4 + 2} = -\frac{1}{7}.$$

Remark. Note that

(a) We did not need to explicitly solve for y to find $\frac{dy}{dx}$.

(b) We have seen that when an equation defines y implicitly in terms of x, there may be multiple explicit solutions for y, or even none. Since $\frac{dy}{dx}$ is given in terms of both x and y, it holds regardless of which explicit solution for y is used, or if there is one at all!

The technique we have just seen is called *implicit differentiation*. Given an equation which defines y implicitly in terms of x, implicit differentiation is the following two-step process:

(1) Differentiate both sides of the equation, using the chain rule each time one differentiates y.

(2) Solve the equation for $\frac{dy}{dx}$ in terms of x and y.

3.) If $x^2 + y^2 = 25$,

(a) Find $\frac{dy}{dx}$.

(b) Find the equation of the tangent line to the circle at the point $(3, 4)$.

4.) The equation $x^3 + y^3 = 6xy$ defines a curve called the *folium of Descartes*, which is shown below:

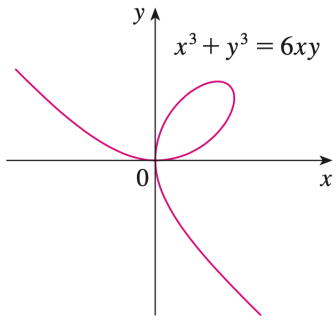


Figure 2: The folium of Descartes.

(a) Find $\frac{dy}{dx}$.

(b) Find the equation of the tangent line at the point $(3,3)$.

(c) At what point in the first quadrant is the tangent line horizontal?

5.) Find y' if $\sin(x+y) = y^2\cos(x)$.

6.) Use implicit differentiation to prove the Power Rule $\frac{d}{dx}x^n = nx^{n-1}$ when $n = \frac{p}{q}$ is a rational number. (*Hint*: If $y = x^{\frac{p}{q}}$ then $y^q = x^p$.)

7.) Find $\frac{d^2y}{dx^2}$ if $x^2 + y^2 = 25$.